

Daily Logic Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Compound Implications · Truth Tables & Model Counting · Exactly-One-True Deductions · Implication Chaining.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Rapid formal conditionals: truth values, implication chaining, and model counts.

1. Determine whether the compound statement $((P \Rightarrow Q) \vee (Q \Rightarrow P))$ is a tautology (always true regardless of the truth values of propositions P and Q).
2. If the conditional statement $(P \vee Q) \Rightarrow (R \wedge S)$ is false, and R is known to be true, what is the truth value of proposition S ?
3. Given that $P \Rightarrow Q$ and $Q \Rightarrow R$ are both true, but R is false, what are the truth values of P and Q ?
4. For three independent propositions P, Q, R , for how many of the 8 possible truth assignments is the implication $(P \wedge Q) \Rightarrow R$ false?
5. For three independent propositions P, Q, R , for how many of the 8 possible truth assignments is the implication $(P \vee Q) \Rightarrow R$ false?
6. Given propositions P and Q , if $P \Rightarrow Q$ is true and $Q \Rightarrow P$ is false, what is the truth value of the biconditional $P \iff Q$?
7. True or false: The statement $P \Rightarrow (Q \Rightarrow R)$ is logically equivalent to $(P \wedge Q) \Rightarrow R$.
8. State the negation of: “If the polynomial $f(x)$ has a root at $x = 1$, then $f(1) = 0$ and $f'(1) = 0$.”
9. If $P \Rightarrow Q$, $Q \Rightarrow R$, and $R \Rightarrow P$ are all true, what can you deduce about the truth values of P, Q, R ?
10. Consider the four statements: (1) $P \vee Q$, (2) $P \Rightarrow R$, (3) $Q \Rightarrow R$, and (4) $\neg R$. Is this set of four statements consistent (can they all be simultaneously true)?

Section B Manipulation Drills

(≤50 s each)

Urns, liars, compound deduction, and truth-table model counting.

1. Three statements about an integer n are given:

- I. n is divisible by 4.
- II. n is odd.
- III. n is divisible by 2.

Given that **exactly one** of I, II, III is true, which statement is impossible to be the true one?
(after TMUA 2016 Paper 2 Q4)

2. Consider three propositions P, Q, R and the implications $S_1 : P \Rightarrow Q$, $S_2 : Q \Rightarrow R$, and $S_3 : R \Rightarrow P$. Given that **exactly two** of S_1, S_2, S_3 are true, can P, Q, R all have the same truth value? (after TMUA 2020 Paper 2 Q20)

3. Three urns A, B, C each have a label:

Urn A : "The prize is in this urn."

Urn B : "The prize is not in Urn A ."

Urn C : "The prize is not in this urn."

Given that the prize is in exactly one urn and **exactly one** label is true, which urn contains the prize?

4. For three propositions P, Q, R , determine the number of truth assignments (out of 8) for which $((P \Rightarrow Q) \Rightarrow R)$ is true.

5. Three suspects X, Y, Z are questioned about a crime committed by exactly one person:

X : "Y did it."

Y : "Z did not do it."

Z : "I did not do it."

Given that the guilty person is the only liar (the two innocent suspects tell the truth), who committed the crime? (after TMUA 2017 Paper 2 Q2)

6. Four statements about a positive integer n are given:

S_1 : $n > 2$ is prime.

S_2 : n is odd.

S_3 : n is a multiple of 3.

S_4 : $n = 9$.

Given that **exactly three** of these statements are true, determine the unique value of n .

7. Given propositions P and Q , if $(P \wedge \neg Q) \vee (\neg P \wedge Q)$ is false and $P \vee Q$ is true, determine the truth values of P and Q .

8. Prove by logical equivalence that $(P \Rightarrow R) \wedge (Q \Rightarrow R)$ is equivalent to $(P \vee Q) \Rightarrow R$.

9. For three propositions P, Q, R , for how many of the 8 possible truth assignments is $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ true?
10. On an island where Knights always tell the truth and Knaves always lie, inhabitant A says: “At least one of us is a Knave.” Inhabitant B makes no statement. What are the identities of A and B ?

Section C Substitution & Structure(≤ 90 s each)*Authentic TMUA Paper 2 Arg1 multiple-choice questions.*

1. For non-zero real numbers x, y , which one of the following is sufficient to conclude that $x < y$?
(after TMUA 2016 Paper 2 Q10)
- A) $x^2 < y^2$
 - B) $y^2 < x^2$
 - C) $|x| < |y|$
 - D) $|y| < |x|$
 - E) $x^5 < y^5$
 - F) $y^5 < x^5$
2. Let P, Q, R, S be four propositions. For how many of the 16 possible truth assignments of (P, Q, R, S) is the compound statement $(P \Rightarrow Q) \Rightarrow (R \Rightarrow S)$ false? (after TMUA 2019 Paper 2 Q3)
- A) 1
 - B) 2
 - C) 3
 - D) 4
 - E) 6
3. Let $f(x) = x^2 + bx + c$ with $b, c \in \mathbb{R}$. Consider three statements:
- I. $f(x) = 0$ has two distinct real roots.
 - II. $c < 0$.
 - III. $b^2 - 4c < 0$.
- Which of the following describes the possible numbers of true statements among I, II, III? (after TMUA 2020 Paper 2 Q12)
- A) Exactly 1 only
 - B) Exactly 2 only
 - C) Exactly 1 or exactly 2 only
 - D) Exactly 0, 1, or 2
 - E) Any number from 0 to 3

4. Consider propositions P, Q, R . You are given that $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ is false. Which of the following CANNOT be the truth assignment (P, Q, R) ? (after TMUA 2017 Paper 2 Q7)

- A) (T, F, T)
- B) (T, F, F)
- C) (T, T, F)
- D) (F, T, F)
- E) (F, F, F)

5. Five logicians each make a statement, as follows:

Mr P: "Of these five statements, an odd number are true."

Ms Q: "Both statements made by women are true."

Mr R: "My first name is Robert and Mr P's statement is true."

Ms S: "Exactly one statement made by a man is true."

Mr T: "Neither statement made by a woman is true."

How many of the five statements can be simultaneously true? (after TMUA Specimen Paper 2 Q20)

- A) none
- B) 1 only
- C) 2 only
- D) 3 only
- E) 4 only

6. In the Cartesian plane, let L be a line passing through $(1, 2)$. Let P be the statement: "If the y -intercept of L is negative, then the x -intercept of L is positive." Which of the following must be true? (after TMUA 2022 Paper 2 Q5)

- I. P
 - II. The converse of P
 - III. The contrapositive of P
- A) none of them
 - B) I only
 - C) II only
 - D) III only
 - E) I and II only
 - F) I and III only
 - G) II and III only
 - H) I, II and III

7. Propositions P and Q are given. Which of the following statements is logically equivalent to “ P is necessary and sufficient for Q ”? (after TMUA 2020 Paper 2 Q20)
- A) $\neg P \iff \neg Q$
 - B) $(P \Rightarrow Q) \vee (Q \Rightarrow P)$
 - C) $\neg P \Rightarrow \neg Q$
 - D) $P \vee Q$
8. A theorem states: “For all real numbers x , if x is rational, then x^2 is rational.” Which of the following three restatements of the theorem are correct? (after TMUA 2023 Paper 2 Q11)
- I. For all real numbers x , x^2 is rational only if x is rational.
 - II. For all real numbers x , a sufficient condition for x^2 to be rational is that x is rational.
 - III. For all real numbers x , if x^2 is not rational, then x is not rational.
- A) none of them
 - B) I only
 - C) II only
 - D) III only
 - E) I and II only
 - F) II and III only
 - G) I and III only
 - H) I, II and III

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. A region is defined by the inequalities $x + y > 10$ and $x - y > 2$. Consider the three statements:
I. $x > 6$ II. $y > 4$ III. $(x + y)(x - y) > 20$. Which of the above statements is/are true for every point in the region?
- A) none
 - B) I only
 - C) II only
 - D) III only
 - E) I and II only
 - F) I and III only
 - G) II and III only
 - H) I, II and III
2. Prove by contradiction: “There is no pair of positive integers (m, n) with $m^2 - n^2 = 2018$.”
3. Prove directly: “If $n > 4$ is such that both $n - 1$ and $n + 1$ are prime, then n is a multiple of 6.”

4. Prove directly: “If x and y are positive reals with $x \neq y$, then $\frac{x+y}{2} > \sqrt{xy}$.”
5. Statement T : “If n is a positive integer such that n^2 is a multiple of 6, then n is a multiple of 6.” (a) Is T true? Prove or disprove. (b) Contrast with the fact that “ n^2 is a multiple of 4” does *not* force “ n is a multiple of 4” (witnessed by $n = 6$). What property of 6, that 4 lacks, makes T true?

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths